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Subject Code: BMC052

Roll No:

MCA

(SEM. IV) THEORY EXAMINATION 2025-26

COMPUTER GRAPHICS AND ANIMATION

TIME: 3 HRS

M.MARKS: 70

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.

2x7 = 14

(a)	Define computer graphics and mention any two applications of computer graphics.	CO1	K2
(b)	Differentiate between raster scan display and random scan display.	CO1	K2
(c)	What is the role of the frame buffer and video controller in a graphics system?	CO1	K2
(d)	State the basic idea of the Text clipping.	CO2	K2
(e)	What is a Bezier curve? Mention its use in computer graphics.	CO3	K2
(f)	What is back-face detection? Mention its use in hidden surface removal.	CO4	K2
(g)	List any four principles of animation used in multimedia system design.	CO5	K2

SECTION B

2. Attempt any three of the following:

7 x 3 = 21

(a)	Differentiate between interactive and passive computer graphics with suitable examples.	CO1	K2
(b)	Explain the working of the Bresenham line drawing algorithm with a suitable numerical example.	CO2	K4
(c)	Compare the Cohen-Sutherland and Liang-Barsky line-clipping algorithms, including their respective conditions and advantages.	CO3	K4
(d)	Explain transparency and shadows in computer graphics. Also discuss their role in generating realistic images.	CO4	K3
(e)	Explain the stages of storyboarding and discuss the role of color theory and sketching in multimedia design.	CO5	K3

SECTION C

3. Attempt any one part of the following:

7 x 1 = 7

(a)	Explain the types of computer graphics and describe random scan and raster scan displays with neat diagrams.	CO1	K2
(b)	Apply the midpoint circle algorithm to generate the points of a circle with centre (0, 0) and radius 8. Show the decision parameter values for the first octant.	CO1	K4

4. Attempt any one part of the following:

7 x 1 = 7

(a)	A triangle has vertices A(2, 2), B(6, 2), and C(4, 5). Apply scaling by $S_x = 2$ and $S_y = 3$ followed by translation $T_x = 4$ and $T_y = -2$. Compute the final transformed coordinates using matrix representation.	CO2	K4
(b)	Explain the complete 2-D viewing pipeline. Also, describe Sutherland-Hodgman polygon clipping with a suitable example.	CO2	K4

5. Attempt any one part of the following:

7 x 1 = 7

(a)	Explain 3-D object representation and 3-D geometric primitives. Discuss 3-D transformations and projection techniques used in computer graphics.	CO3	K3
(b)	Describe Bezier and B-spline curves. Explain their properties and importance in curve and surface modelling.	CO3	K2

6. Attempt any one part of the following:

7 x 1 = 7

(a)	Compare the Depth buffer, A-buffer and Scan line methods for hidden surface removal. Discuss their advantages and limitations.	CO4	K4
(b)	Explain illumination models in computer graphics. Discuss ambient light, diffuse reflection, specular reflection, and the Phong model, with suitable expressions.	CO4	K3

7. Attempt any one part of the following:

7 x 1 = 7

(a)	Explain the fundamentals of multimedia system design. Discuss animation techniques, animation file formats and tools used for web animation.	CO5	K3
(b)	Design the sequence of steps for creating a bouncing ball animation. Explain the use of timing, squash and stretch, motion path, GIF, plugins and players in web-based animation.	CO5	K3